

Apis Insight for Phenotype Classification and Hive Health Forecasting using IoT and Deep Learning

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Abstract—This research illuminates the pivotal quandary of bee health classification native to the Indian subcontinent, a crucial facet of maintaining agroecological equilibrium. Through a multifaceted integration of advanced computer vision, intricate deep learning paradigms, and hardware monitoring, the study explores the application of cutting-edge technologies to monitor and assess bee health and subspecies as well as beehive assessment. By incorporating hardware-based monitoring systems, the research exemplifies the vanguard potential of technologically entrenched methodologies in the realm of agricultural sustainability. This approach not only addresses the complexities of bee phenotype classification within the specified geographical context but also emphasizes the utilization of hardware components for real-time environmental data collection, fostering a paradigmatic shift in the axiomatic fulcrum of agro-symbiotic consilience.

Keywords—agro-symbiotic consilience, hardware monitoring systems, computer vision, deep learning, agricultural sustainability

I. INTRODUCTION

Pollinators, with a focus on bees, play a crucial role in preserving agroecological balance by facilitating the pollination of flowering plants, thereby supporting agricultural productivity and biodiversity. In the Indian subcontinent, where agriculture is integral to the economy, ensuring the health and vitality of bee populations is paramount. This research initiative embarks on a multifaceted exploration, addressing key aspects: (1) the pressing challenge of bee health; (2) subspecies identification for both commercial and research applications; (3) beehive health assessment and real-time monitoring; and (4) tackling the significant issue of colony collapse disorder (CCD) in India. The central objective of the study being conducted is the creation of a durable model that can help with the conjecture of the health status, subspecies classification, and beehive health status of individual bees. Through the convergence of technology and biology, this research aims to offer a comprehensive solution that empowers beekeepers, farmers, researchers, and conservationists with real-time insights into bee populations. This, in turn, facilitates informed decision-making and timely interventions to safeguard these indispensable contributors to our ecosystem.

Safeguarding these vital contributors to our ecosystem has been a focal point of recent research endeavors, with image classification algorithms being increasingly employed to monitor bee health and establish beehive monitoring systems. Precision apiculture, denoted as PA, introduces a

pioneering framework for the administration of apicultural settings. This methodology centers on the detailed observation of singular bee colonies as a fundamental aspect of apiary management. This approach entails the systematic observation of individual bee colonies to enhance productivity while concurrently minimizing resource utilization. It is categorized as a subfield within precision agriculture. Precision apiculture aims to enhance efficiency and sustainable practices in beekeeping operations. Previous studies have investigated the utilization of hive weight and delved into live climatic conditions such as internal and external hive temperature, dampness, as well as acoustic signals and flight activity around the beehive to evaluate hive status. [1]. Simultaneously, various researchers have explored the utilization of imagery data and micro-climate parameters to monitor the health of beehives. Nevertheless, prevalent monitoring systems frequently demonstrate elevated energy demands, relying on non-rechargeable batteries that require frequent upkeep. This reliance on maintenance procedures may induce stress among bees, potentially compromising their overall well-being. [2]. Moreover, remote apiaries often lack wall power connections. To overcome these challenges, there is an urgent requirement to devise a methodology capable of harnessing energy from ambient sources, proficiently managing the gathered energy, and ensuring the uninterrupted operation of monitoring systems for honeybees.

In the recent past, the rise of the field of hardware and software integration using the Internet of Things (IoT) in Precision Agriculture has given rise to numerous proposals aimed at advancing the study of Precision agriculture. These proposals vary in technical concepts and associated production costs. The proliferation of IoT systems has been accompanied by the widespread availability of cost-effective Commercial Off-The-Shelf (COTS) components[1][3]. In this context, various remote monitoring systems have been suggested to alleviate the financial burden of apiary maintenance while simultaneously enhancing colony production. Furthermore, these systems facilitate extensive data collection for scientific research and investigative purposes, making a substantial contribution to mitigating the decline in bee colonies and fostering a more profound understanding of the factors influencing beekeeping activity [3].

Assessing the health of a beehive population requires an initial examination of the individual phenotype classification of bees. Recent advancements in Deep Learning (DL) and Image Processing show promising potential for automatically

identifying bees and evaluating their well-being. This project encompasses considerations for environmental conditions such as temperature and humidity, along with the detection of signs indicating Varroa destructor infestation. Varroa destructor, a parasitic mite that infests honeybee colonies, is widely recognized as one of the most significant threats to bee populations globally. While Varroa itself does not directly cause diseases such as Deformed Wing Virus, Acute Bee Paralysis Virus, and Chronic Bee Paralysis Virus, it weakens bees, rendering them more susceptible to various viruses and pathogens, which can have detrimental effects on the growth and survival of bee offspring. This susceptibility leads to a loss of bee life, particularly during the winter season when temperatures plummet, rendering the hive more vulnerable to damage. Leveraging these technological advancements, the classification of a bee's health status can be achieved through comprehensive image analysis.

This research outlines our endeavours to construct a cost-effective and modular system that is designed for easy deployment and upgrades. Our objective is to obtain high-quality data using predominantly consumer-grade equipment. The purpose of this system is to assist beekeepers by delivering pertinent information about the condition of their bees and live beehive status using real-time monitoring [3]. Additionally, it aims to offer warnings concerning unhealthy hive activity or any unhealthy bee either through image analysis or through the identification of predetermined patterns and signatures indicative of bee health issues [1].

The subsequent sections of this paper are organized as follows: Section II presents the Literature Review; Section III provides an overview of the diverse bee species present in the Indian subcontinent. Following that, Section IV elucidates the live beehive monitoring system, while Section V delineates the data utilized for analysis. Sections VI, VII, and VIII explore Honey Bee health, Subspecies, and pollen classification, respectively, while Section IX expounds on the methodology employed for real-time monitoring in the study of beehives. Lastly, Section X encapsulates the principal findings derived from this research.

II. LITERATURE REVIEW

The assessment of bee health and the management of beehive conditions are crucial for sustainable beekeeping practices and the preservation of pollination ecosystems. Various studies have explored innovative methods for monitoring bee health and hive conditions using advanced technologies. Myers [1] discusses the utilization of hive-streaming data analytics to measure beehive health, emphasizing the importance of real-time data insights for the early detection of abnormalities. Similarly, Braga et al. [2] present an intelligent monitoring system that leverages IoT technology for assessing bee hive health, demonstrating the potential of IoT-based solutions in beekeeping management. Additionally, Cota et al. [3] propose Bee Hive Sense, an integrated information system architecture for sustainable remote monitoring and management of apiaries, showcasing the benefits of microservices and IoT integration in beekeeping operations.

In parallel, advancements in machine learning and deep learning have revolutionized bee health assessment and beehive monitoring. Zaman and Dorin [8] introduce a framework for sensor-based beehive health monitoring,

highlighting the significance of machine learning algorithms in processing sensor data and extracting meaningful insights. Edwards-Murphy et al. [10] present b+WSN, a smart beehive system incorporating preliminary decision tree analysis, demonstrating the potential of machine learning techniques for agriculture and honey bee health monitoring. Furthermore, recent studies such as Kelley et al. [9] and Lv et al. [15] explore deep learning models for image classification and bee species classification, underscoring the role of neural networks in identifying bee health indicators and subspecies classification. Collectively, these studies underscore the multidisciplinary approaches and technological innovations driving advancements in bee health assessment and beehive monitoring, paving the way for sustainable beekeeping practices and ecosystem conservation.

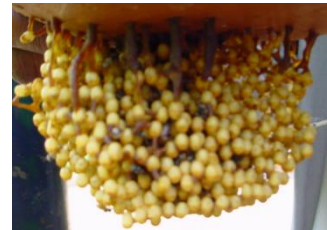
III. HONEY BEE SPECIES AND BEEHIVE CLASSIFICATION

A. Dammer Bee(*Tetragonula iridipennis*)

The Indian stingless bee, scientifically known as *Tetragonula iridipennis*, is a bee species within the Apidae family and Apinae subfamily.



Fig. 1.



Dammer Bee and its Beehive

Fig. 1. demonstrates the Indian stingless bee, *Tetragonula iridipennis*, requires an optimal hive The temperature varies between thirty-four point five and thirty-five point five degrees Celsius [4]

B. Italian Honey Bee

The initial introduction of domesticated honeybees to North America involved the Italian honeybee species (*Apis mellifera ligustica*). These bees are known for their amiable nature and extensive distribution, these honeybees have achieved widespread prevalence, primarily due to their remarkable adaptability to diverse climates. Characterized by yellow-brown and dark brown stripes on their bodies, these bees demonstrate swift reproductive capabilities and sustain substantial colonies even in the winter months [5].



Fig. 2. Italian honey bee and its beehive

Fig. 2. shows the Italian Honey Bee. *Apis mellifera* employs both heating and cooling behaviors to regulate the nest temperature, aiming to maintain stability close to the ideal temperature of 35°C. This encompasses a range of 32–36°C.

C. VSH Italian Honey Bee

VSH, or Varroa Sensitive Hygiene, Italian bees are a type of genetically modified honeybees that allows a colony to endure without the need for mite controls as shown in Fig. 3. This trait induces VSH hygienic behavior, involving the removal of cells infested with mites from the brood nest, within a beehive.



Fig. 3. VSH Italian honey bee and its beehive

Honey bee hives typically maintain a temperature range between 33–36°C, coupled with an average humidity level of around 52% [6].

D. Indian Hive Bee

The mentioned species is *Apis cerana indica*, commonly recognized as the Asiatic honey bee or Eastern honey bee as shown in Fig. 4. Originating in India and Asia, these bees play a crucial role in the commercial production of honey and other bee-derived products. Acknowledged for their significance as pollinators for coconut palms, these bees are prone to absconding and swarming behaviors.



Fig. 4. Indian hive bee and its beehive

IV. BEEHIVE MONITORING SYSTEM USING IOT

In recent years, as the global COVID-19 pandemic has disrupted the supply and logistic food chains worldwide, the agricultural sector has had to face numerous problems on a worldwide level [2]. Various problems like the increase in global warming levels, climatic changes, and frequent weather disruptions and storms have added to these difficulties. Hence incorporating new real-time monitoring systems and using new technological solutions, there is potential for humanity to overcome these predicaments by

addressing the intricate relationships within the food-water-energy-climate nexus [7]. This opens avenues for the transformation of agriculture from traditional practices to a digital paradigm.

The real-time monitoring system is employed for overseeing sensors to assess the internal conditions of the beehive. The temperature and humidity observations were captured and showcased on a 16x16 LCD as well as on the web interface, connected to the circuit board through an I2C module. Subsequently, the recorded data was uploaded to the Arduino Uno Board for processing. After converting the binary data into numerical format, it was transmitted through the COM9 serial port connection, and the outcomes were exhibited on the Arduino Serial Port. The code for the temperature sensors utilized the Arduino DHT11 library. As an additional enhancement to the monitoring system, the data from the COM9 Serial Port was routed through the Flask module to the designed web interface, allowing the recorded values to be displayed on the website.

The comprehensive setup involves a range of key components, including a DHT11 sensor, a 16x16 LCD module, an I2C module, a 220 Ohm ¼ Watt resistor, an electronic breadboard, an Arduino UNO Board, B10 Buzzers, and carefully connected jumper cables. The diligent DHT11 sensor continuously monitors the intricate balance of temperature and humidity conditions within the beehive. In the event of substantial temperature and humidity fluctuations, the sensor promptly triggers buzzer warnings, ensuring swift alerts to both vigilant beekeepers and dedicated researchers as shown in Fig. 5.

An essential advantage of monitoring beehives lies in the timely identification of potential issues. Instances like abrupt fluctuations in temperature or humidity signal potential stress within the hive, enabling swift intervention to pinpoint and resolve underlying problems like diseases or pest infestations [8]. Monitoring the hive's weight is a crucial aspect of beekeeping that offers valuable insights into honey production cycles. By regularly assessing the hive's weight, beekeepers can effectively gauge the colony's productivity and health. This practice becomes particularly significant in determining the opportune moment for harvesting honey, striking a balance between maximizing yield for beekeepers and ensuring there is a sufficient honey reserve left for the sustenance of the hive. The precision in timing that comes from monitoring hive weight contributes to the overall well-being of the bee colony and supports sustainable beekeeping practices. This approach aligns with the principles of responsible hive management, ensuring both the welfare of the bees and the efficiency of honey extraction processes.

The proposed setup continuously monitors the health of a beehive by observing the bees' stress behavior, particularly during winter seasons. As temperatures drop and environmental conditions become harsher, bees may display stress behaviors such as clustering together to generate metabolic warmth or adjusting humidity levels by transporting water into the hive [9]. By constantly monitoring temperature and humidity levels in real time, the IoT model can detect these stress responses and alert beekeepers to take

necessary actions. The setup includes buzzer warnings triggered by significant fluctuations in temperature or humidity. These warnings serve as immediate alerts to beekeepers, prompting them to promptly investigate and address potential stressors within the hive. Furthermore, the data collected by the IoT system is displayed in real-time on a web interface. This feature enables beekeepers to remotely monitor hive conditions and intervene in response to stress behaviors observed during winter seasons, even if they are not physically present at the hive-site.

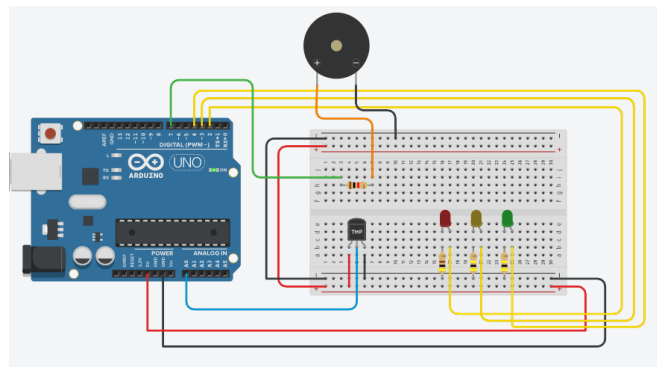


Fig. 5. Live Monitoring Setup

This above framework is based on Real-time Performance Monitoring of the beehive. This involves the regular collection of live beehive conditions data to create a record of the status of a bee colony at any moment. Performance monitoring abstains from drawing conclusions or offering interpretations of the monitored data.

Temperature: The reproductive process of honeybees requires a consistent and moderate air temperature within the hive, ideally ranging between 14.28 and 36.5°C as shown in Fig. 6. To sustain the right environmental conditions inside the hives, bees utilize adaptive techniques. In cases of lower temperatures, they produce metabolic warmth by clustering together [10][11]. Conversely, in the case of elevated temperatures, they transport water into the hive, thereby augmenting humidity.

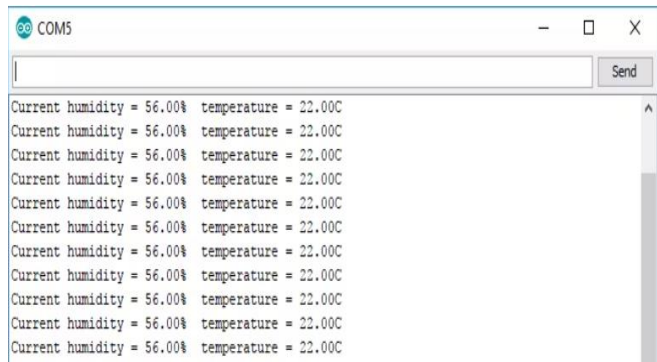


Fig. 6. Serial Monitor Output

Humidity: Assessing the relative humidity within beehives stands as a crucial factor in gauging the condition of colonies and plays a direct role in influencing productivity [11]. Existing studies affirm that temperature and humidity

levels exhibit variations based on the subspecies of bees. Maintaining a relative humidity within the hive ranging from 90% to 95% is essential for optimizing the hatching of eggs as shown in Fig. 6.

V. DATA DESCRIPTION

This section delves into the comprehensive examination of multiple attributes essential for mapping and providing an in-depth analysis of Bee phenotype classification. The dataset under consideration encompasses 1795 images, showcasing diverse Indian bee subspecies, including the Dammer bee, Italian bee, VSH Italian bee, and Indian hive bee. The data collection and comparisons primarily originate from various locations such as Kalakadu, Nedumangad, Almora, Bela Greya, and Alchuna.

Moreover, for reference and sub-species comparison, we have utilized the BeeImage Dataset: Annotated Honey Bee Images [12]. This dataset serves as a valuable resource in augmenting our understanding and analysis of different bee subspecies.

Table 1. Attributes Classification

Attributes	Values
Location	Kalakadu, Bela Greya, Nedumangad, Almora, Alchuna, Adilabad
Subspecies	Dammer bee, Italian Honey bee, VSH Italian Honey Bee, Indian Hive Bee
Health	Health, Unhealthy, Few varroa mites
Pollen-type	Eucalyptus, Mustard, Sunflower, Turnip and Coriander
Caste	Queen, Worker, and Drone

Table I provides a comprehensive depiction of the various values (classes) corresponding to each attribute within the dataset problem. It is worth noting that specific attributes, particularly subspecies, are allocated for instances where examples remain unidentified. The meticulous annotation of the health attribute was carried out by seasoned field experts during the image capture of honey bees, ensuring the accuracy and reliability of the dataset.

Furthermore, the health attribute extends its coverage to encompass hive-related issues, offering detailed categorizations such as "Healthy," "Unhealthy," and "few varroa mites." This nuanced classification provides a more intricate understanding of the well-being status of honey bee data in the dataset. Additionally, the dataset incorporates information about the type of pollen, a crucial element for determining the optimal crop selection in the nearby vicinity of the beehive to facilitate optimal cross-pollination. This deliberate inclusion holds significance as it recognizes that changes in the beehive's condition can serve as indicators of alterations in honey bees and vice versa, creating a robust and interconnected dataset for analysis and classification purposes.

VI. HEALTHY STATUS CATEGORIZATION

For classification of the health status of the bee which involves an image dataset, the data comprises cropped images of honey bees, each associated with various attributes. These attributes signify the current health status of the beehive the honey bee's subspecies and its current involvement in pollination. In the development of the classifier, emphasis was placed on the health attribute, serving as a direct indicator of the bee's well-being [10]. Moreover, classifier development considered distinct aspects, including the attributes related to subspecies and the transportation of pollen, as they contribute to a more nuanced understanding of hive health. Examining the pollen-carrying attribute, for instance, allows for assessing the availability of food supplies by counting the number of bees transporting pollen [3]. Meanwhile, subspecies information aids in identifying potential hive-robbing instances, where bees from another hive, possibly of a different species, invade for nectar and honey collection. These attributes collectively demonstrate their potential to contribute to the classification of bee health [2]. Furthermore, the aspect of pollen carrying can be further studied to specify the type of pollen that is optimal for the beehive to transport.

VII. SUBSPECIES CLASSIFICATION

In the context of subspecies classification, the various Indian subspecies like (*Trigona iridipennis*) Dammer bee, (*Apis Mellifera*) Italian bee, VSH Italian Honey Bee, and (*Apis cerana indica*) Indian Hive bee were considered. The subspecies classification in this research is accomplished through a well-designed deep-learning model. Leveraging Convolutional Neural Networks (CNNs), the model processes intricate visual patterns and distinctive features present in bee images to classify them into distinct subspecies. Through convolutional layers, the network learns hierarchical representations of the input data, enabling it to discern subtle differences in morphology and coloration that characterize various bee subspecies [13]. The utilization of deep learning techniques facilitates automated and accurate subspecies identification, contributing to the broader understanding of bee biodiversity. This approach not only streamlines the classification process but also establishes a foundation for monitoring and preserving diverse bee populations, thereby fostering sustainable agricultural practices and environmental conservation [8].

VIII. POLLEN CLASSIFICATION

The act of pollination holds a crucial function as a regulatory ecosystem service in the natural environment. Within the realm of insect pollinators, the predominant contributors to pollination in both managed and natural ecosystems are solitary and social bees.

The major pollinating plants recognized for various bee species are:

- Dammer bee: This bee primarily visits crops exhibiting extensive blooming patterns, such as coffee, guava, mango, and Eucalyptus with

Eucalyptus being the primary seed being pollinated [4].

- Italian honey bee: This bee visits crops like Mustard, Sunflower, Cotton, and Apple.
- VSH Italian honey bee: This bee is a genetically modified version of an Italian bee with resistance against Varroa Mites and primarily pollinates Sunflower and Mustard.
- Indian hive bee: This bee primarily pollinates Turnip, melon, pumpkin, radish, and coriander.

IX. PROPOSED METHODOLOGY

The goal of the Model was to correctly assess the bee health and subspecies based on the variety of attributes. For comparison with other models 2 models were narrowed down for the bee image processing to correctly determine the correct bee phenotype. In the provided code, a convolutional neural network (CNN) based on the MobileNetV2 architecture is used for bee health classification. The MobileNetV2 is designed for mobile and embedded vision applications, it is a lightweight and efficient deep learning model that was proposed for use in remote environments, where most beehives and apiaries are located. It is an extension of the original MobileNet architecture, incorporating inverted residual blocks with linear bottlenecks to enhance feature extraction while maintaining computational efficiency. During the implementation phase, many challenges were encountered with suboptimal model performance metrics, prompting iterative adjustments to enhance the overall system. Initially utilizing a Sequential Model, we transitioned to a transfer learning approach, employing the MobileNetV2 architecture for improved feature extraction [14]. Methods for augmenting data were employed for addressing limited training data, and the model underwent fine-tuning to enhance performance. The model architecture was serialized into a JSON file, and class labels were saved separately for reproducibility. These modifications, guided by experimentation and evaluation metrics, ultimately culminated in an enhanced system poised for reliable and accurate bee health assessment, showcasing the adaptability and resilience of the proposed method.

To improve the model's accuracy and F1 score, several strategic modifications were implemented. Initially, the Sequential Model architecture was replaced with the more sophisticated MobileNetV2, a transfer learning approach known for exceptional attribute extraction capabilities. By leveraging the pre-trained weights from the provided dataset, the model gained a nuanced understanding of intricate features within bee images, contributing to improved classification accuracy. Methods for augmenting data were implemented during the training phase to artificially diversify the dataset, reducing and mitigating overfitting while improving the model's ability to generalize to unseen data. Additionally, hyperparameters were fine-tuned, including adjustments to learning rates and the introduction of dropout layers to mitigate potential overfitting. The implementation

of early stopping and model checkpoint callbacks ensured the model's training was optimized, preventing unnecessary epochs, and preserving the best-performing model. Serialization of the model architecture into a JSON file and saving class labels separately streamlined reproducibility and model deployment. These collective modifications reflect a systematic and iterative approach aimed at refining the deep learning model, resulting in a significant improvement in accuracy and F1 score for robust bee health assessment.

In the DL model, several mechanisms are employed to enhance accuracy by mitigating false detections. Data augmentation techniques, such as rotation, zooming, shifting, and flipping, diversify the training data to reduce overfitting and enhance generalization. Early stopping prevents overfitting by halting training when validation metrics plateau, while dropout layers randomly deactivate neurons to improve model robustness. Model checkpointing saves optimal weights, ensuring the restoration of the best-performing model. Additionally, hyperparameter tuning optimizes model performance by systematically adjusting learning rates, batch sizes, and optimization algorithms. Together, these mechanisms work to minimize false detections and improve overall accuracy [15].

The deep learning model accepts image inputs in common formats such as JPEG and PNG, utilizing the `skimage.io.imread()` function. Images are standardized to dimensions of 128x128 pixels with 3 color channels (RGB) using `skimage.transform.resize()`, ensuring consistency across inputs. Resizing is performed with the 'reflect' mode to handle boundaries effectively. Normalization scales pixel values to the range [0, 1] by dividing by 255, enhancing convergence and stability during training.

Otsu's thresholding method is employed for image segmentation, automatically computing a threshold value based on the image histogram to separate foreground and background intensities [16]. This produces a binary image where foreground (bee) regions are white (value 1) and background regions are black (value 0). Contours are found using `skimage.measure.find_contours()`, with the largest contour selected as the region of interest (ROI), indicating significant objects or regions within the image. Bounding box coordinates of the ROI are determined to define the area encompassing it, facilitating subsequent cropping. Otsu's thresholding is advantageous for bee health assessment due to its automatic threshold calculation, maximizing inter-class variance to distinguish bees from the background. Its robustness to varying lighting conditions ensures reliable performance in diverse environments, while its computational efficiency enables real-time processing, ideal for on-site monitoring of bee colonies without manual parameter adjustments.

- **Learning Rate:** The learning rate is implicitly set by the Adam optimizer.
- **Epochs:** The training process is configured to run for a maximum of 200 epochs, determining the number of

passes the model makes through the entire training dataset.

- **Batch Size:** A batch size of 256 is specified, defining the number of training examples utilized in one iteration during the model training process.

The optimization process, the learning rate of the model, and the effectiveness of training are significantly impacted by these hyperparameters. They play a crucial role in controlling the performance of the deep learning model as shown in Fig. 7. The layers employed in the model include:

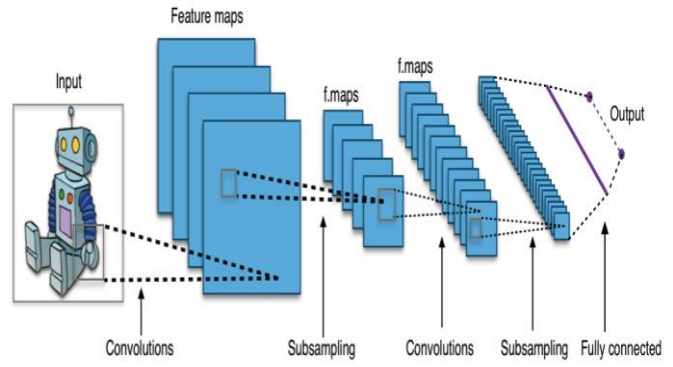


Fig. 7. Model Architecture of CNN-Based Sequential Model

- **MobileNetV2 Base Model:** MobileNetV2 is used as the base model, a pre-trained convolutional neural network (CNN) with efficient inverted residual blocks for feature extraction in image classification.
- **Global Mean Pooling Layer:** The integration of a Global Mean Pooling layer involves the spatial averaging of feature maps obtained from the base model, leading to the reduction of their dimensions to a singular value per channel to facilitate information aggregation.
- **Dense Layer with SoftMax Activation:** The dense fully connected layer serves as the output layer with neurons equal to the number of classes. SoftMax activation produces probability distributions for multi-class classification, determining the predicted bee health category.

Table 2. Performance Metrics

METRICS	CNN-Based Sequential Model	MobileNet V2
Accuracy	0.959731543624161	0.83221476510067
Precision	0.775327452552793	0.70325654901585
Recall	0.96896280217909	0.76214887640449

F1-Score	0.8247126436781	0.7225225225225
	6	2

Furthermore, the original Sequential model is a conventional deep learning architecture where layers are added sequentially. It requires manual configuration of layers, which can be time-consuming and may not be optimized for specific tasks. The layers used for classification were Convolution, Max Pooling, Dropout1, Convolution, Dropout 2, and Dense layer with SoftMax activation [17]. The Sequential Model may be suitable for general image classification tasks but may lack the efficiency needed for real-time applications on resource-constrained devices while the MobileNet V2 is well-suited for the bee health assessment project as it balances accuracy and computational efficiency and is especially useful for on-site monitoring. Table 2 demonstrates the performance metrics obtained by testing the various models.

X. CONCLUSION

In conclusion, the bee phenotype assessment project represents a multifaceted approach leveraging cutting-edge technologies to monitor and evaluate the well-being of bee colonies. Through the integration of image processing, deep learning, and real-time environmental sensing with Arduino modules, our system provides a comprehensive analysis of both individual bee health and overall hive conditions.

The implementation of various models like CNN (Based on Sequential Architecture), and MobileNetV2-based deep learning model for bee health classification demonstrates the project's commitment to accuracy and efficiency in image analysis. Despite challenges in optimizing the model, iterative modifications, including data augmentation, increased training epochs, and changes in model architecture, have been undertaken to enhance predictive performance [14].

The live gathering and analyzing of hive climatic conditions using Arduino modules enriches the project's scope, offering valuable insights into environmental factors affecting bee colonies. Additionally, the incorporation of a web-based interface facilitates user-friendly interactions, enabling beekeepers and researchers to upload images for bee and hive health assessments seamlessly. Fig. 8. describes the architecture diagram for the project.

While MobileNetV2 is well-suited for resource-constrained scenarios and image classification tasks, there might be specific cases where using a Sequential Model could offer advantages as the Sequential Model allows for greater customization of layers and architecture, making it more adaptable to specific requirements of the bee phenotype assessment task. Customizing the model architecture may be beneficial for fine-tuning the model to the unique features and nuances of bee images. As the dataset for bee phenotype classification is large and domain-specific, a Sequential Model can be trained from scratch or fine-tuned more effectively compared to a pre-trained MobileNetV2, and transfer learning using MobileNetV2 might not always be

advantageous if the source domain (ImageNet) significantly differs from the target domain (see images).

On training and testing the various models, The Sequential Model provides a clearer and more interpretable structure, making it easier to debug and understand the model's behavior during training and inference while the MobileNetV2's complex architecture may pose challenges in terms of interpretability, especially for users who require a detailed understanding of the model with the only demerit of the Sequential Model being the lack of computational resources [17].

While the project has made significant strides in technological innovation, there remain avenues for future enhancements. Further refinements in the deep learning model, exploration of additional environmental parameters, and integration of automated hive inspection mechanisms could contribute to the project's continuous evolution. Overall, this initiative holds the promise of advancing beekeeping practices, contributing to ecological sustainability, and fostering the well-being of vital pollinators.

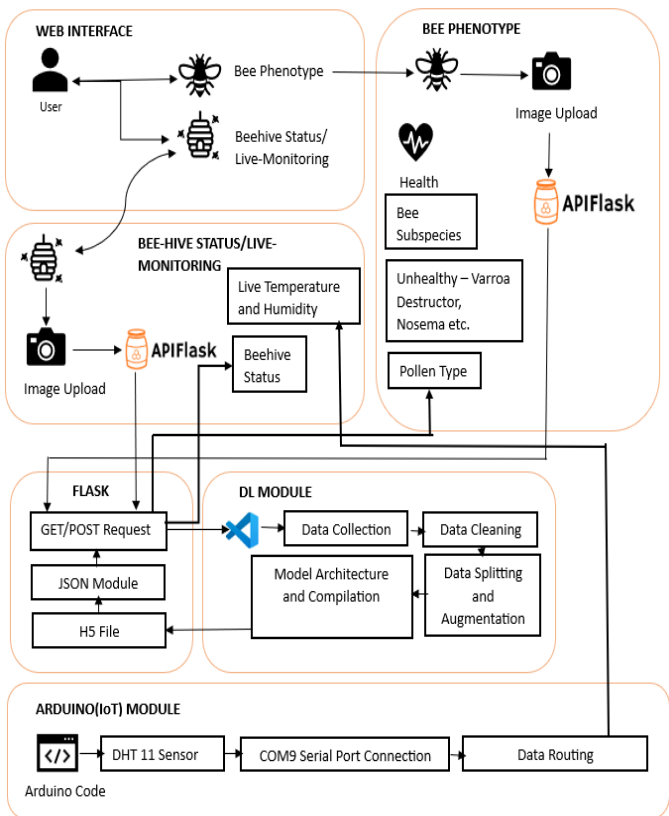


Fig. 8. Architecture Diagram

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